

Marshall Saltz

EDUCATION

Honors Bachelor of Electrical and Computer Engineering with Minor in Computer Science

Sep. 2021 - Jun. 2025

Oregon State University, Corvallis, OR

GPA: 3.25

TECHNICAL EXPERIENCE

Student Researcher – Robotics Lab

Oregon State University, Corvallis, OR

Jan. 2022 - Jun. 2025

- Designed and tested over 200,000 simulated gripper configurations to optimize reachability using evolutionary algorithms.
- Verified results in PyBullet simulation using Python and inverse kinematics to be published in undergraduate thesis paper, improving on previous research completed.
- Developed effective technical documentation and analyzed results for honors thesis in an R&D environment, ensuring effective transfer of information for future project.

Intern – Artificial Intelligence in Agriculture

Jun. 2023 - Aug. 2023

AgAID, Corvallis, OR

- Developed Python algorithms integrating RGB and depth sensor data for 3D modeling of trees.
- Delivered a final presentation of research outcomes to a panel of industry leaders and academic professionals, received positive feedback.
- Advanced AI-driven agricultural solutions for testing a robotic system in simulation by contributing to a collaborative codebase, improving simulation performance.

Student Technician – Service Desk

Oregon State University, Corvallis, OR

Nov. 2021 – Jun. 2025

- Diagnosed and resolved hardware/software issues on MacOS/Windows platforms with above a 90% first-contact resolution rate.
- Provided excellent customer service by utilizing strong phone etiquette and conflict resolution strategies.

LEADERSHIP EXPERIENCE

Engineering Student Council EECS Senator

Mar. 2022 - May 2023

Oregon State University, Corvallis, OR

- Represented the EECS student body, facilitating feedback between students and administration.
- Improved student resource accessibility through organized outreach events.

Computer Science Club President

Aug. 2020 - May 2021

Front Range Community College, Fort Collins, CO

- Led the organization of technical workshops, presentations, and a 30+ person hackathon.
- Maintained club engagement during virtual isolation.

PROJECTS

- **Custom Drone:** Senior design team project involving a PCB designed in KiCAD, a machined frame, and a custom controller to build a fully functional drone with auto landing.
- **Robotic Drawing Arm:** Junior design team project involving a custom PCB, Python GUI interface, computer vision, and a chassis designed in Fusion360 and 3D printed to construct a robotic arm that accepts positional commands and replicates images.

SKILLS

- **Electrical:** KiCAD, PCB Design, Flexible and Printed Electronics, VLSI, FPGA, LTSpice, Soldering, Electrical Debugging, AVR Assembly
- **Mechanical:** Fusion360, Power tools, Prototype assembly, Hand tools, 3D printing, AutoCAD
- **Software:** PyBullet, Python, C/C++, MATLAB, Microcontrollers, Linux, Computer Vision
- **Soft:** Technical Writing, Interdisciplinary Collaboration, Problem Solving, Public Speaking, Customer Service, Communication